Samples, Studies, and Bias

Name _____

A radio station asks listeners to call in: *Should school start later?* 90% of callers say yes. Does that mean 90% of all students agree? Not necessarily.

	Meaning	**Example**
Population	The **whole group** we want to learn about.	All 1,200 students at a school.
Sample	The part of the population we collect data from.	100 of those students.

We can only **generalize** (say the results are true for the population) if the sample represents the population well.

Three types of studies

Sample survey	Ask a sample questions.	Ask 100 students how many hours they sleep.
Observational study	Watch or measure. Do **not** change anything.	Compare the grades of students who do and do not play sports.
Experiment	**Do something** to the subjects (a treatment) and measure what happens.	Give half the students a 10-minute break before a test. Compare scores.

Bias is when the way we collect data favors some answers. A biased sample does not represent the population.

Type of bias	**What happens**	**Example**
Voluntary response	People choose if they answer. People with strong feelings answer most.	The radio call-in poll.
Convenience	The sample is whoever is easy to reach.	Asking only your friends.
Undercoverage	Some groups are left out of the sample.	Calling only home phones (many people do not have one).
Nonresponse	People who were chosen do not answer.	Half of the mailed surveys never come back.
Response	The question or the setting pushes people to answer a certain way.	“Don’t you agree that our great principal deserves a raise?”

Work with a partner. Compare answers when you finish each problem.

1. Name the population and the sample.

   (a) A city wants to know how residents feel about a new park. It surveys 400 residents.

   (b) A farmer tests 50 apples from her orchard to check for worms.

   (c) A school picks 30 seniors to ask about graduation plans.

2. Is each study a sample survey, an observational study, or an experiment? Explain.

   (a) Researchers ask 500 adults how many hours of TV they watch.

   (b) A coach has half the team stretch before practice and half not stretch. She counts injuries.

   (c) A scientist records the weights of babies whose mothers smoked and babies whose mothers did not.

   (d) Students give one plant fertilizer and another plant only water. They measure growth.

3. Name the type of bias. Explain how it could change the results.

   (a) A website asks visitors to vote on their favorite phone.

   (b) To learn about cafeteria food, a student asks people standing in the lunch line.

   (c) A survey about internet use is done only on the internet.

   (d) A survey asks, “Do you agree that dangerous skateboarding should be banned from parks?”

4. To find the favorite sport of students at a school, Jamal surveys 60 fans at a basketball game. Can he generalize his results to all students at the school? Explain.

5. Rewrite this question so it does not push people toward an answer:

   “Don’t you think the school day is too long?”